SUVANSU PAUDEL

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PROFESSIONAL SUMMARY

Artificial Intelligence & Machine Learning postgraduate student with 3+ years of software engineering experience and hands-on work in data analysis, statistical methods, machine learning, computer vision, NLP, and generative AI. Experienced with Python, SQL/MySQL, Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, OpenCV, YOLO, Matplotlib, and Seaborn. Strong background in data preprocessing, feature engineering, model evaluation, visualization, algorithm implementation, and production-oriented software development. Experienced collaborating in Agile/Scrum teams, debugging production issues, integrating APIs, and translating technical problems into practical solutions.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MySQL, Java, JavaScript, Progress 4GL, Pandas, NumPy, Jupyter
Statistics & Machine Learning: Statistical Analysis, Classification, Regression, Decision Trees, Random Forest, K-Means, PCA, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning
AI / Deep Learning / Computer Vision: PyTorch, TensorFlow, YOLO, OpenCV, Ultralytics, TorchVision, NLP, RAG, LLMs, LangChain
Data Visualization: Matplotlib, Seaborn, Exploratory Data Analysis, Pattern & Correlation Analysis, Reporting
Data & ML Platforms: Databricks, Databricks Vector Search, Databricks Model Serving, MLflow, Vercel
Software Engineering: REST APIs, Spring Boot, Git/GitHub, CI/CD, Integration Testing, Debugging, API Troubleshooting, Code Reviews, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer | *Javra Software Nepal* | *2021 – Dec 2025*

- Developed client-requested software features using Progress 4GL and JavaScript, applying structured problem solving and maintainable implementation practices.
- Built and supported data-driven Excel and PDF reporting tools based on search queries, transforming application data into actionable reports for users.
- Used Python scripting for internal automation and data processing, supporting repeatable workflows and operational efficiency.
- Assisted in developing monitoring automation for client web pages using Progress 4GL, AJAX, JavaScript, and jQuery to validate application behaviour and identify issues.
- Applied Spring Boot and REST/API integration concepts in backend development and troubleshooting.
- Performed integration testing, investigated production and API/backend issues, resolved client-reported defects, and participated in code reviews within Agile/Scrum workflows.
- Architected and developed an internal Learning Management System used company-wide, including workflow and data-management functionality.
- Implemented an Email Unsubscribe feature with secure backend handling and user preference management.

EDUCATION

Postgraduate Studies in Artificial Intelligence & Machine Learning | *Lambton College, Ottawa, Canada* | *2026 – Present*
Relevant focus: Data Science, Machine Learning, Big Data Analysis, Generative AI, RAG, Python-based analytics and ML workflows.

Bachelor of Technology (B.Tech) in Computer Science and Engineering | *MIT World Peace University, Pune, India* | *2018 – 2022*
CGPA: 8.35 / 10

SELECTED DATA SCIENCE & MACHINE LEARNING PROJECTS

Real-Time Weapon Detection System | Python, YOLO, PyTorch, OpenCV, Ultralytics

- Developed a real-time computer vision system using YOLO object detection to identify weapons in CCTV/video footage.
- Prepared and annotated a custom dataset, automated train/validation splitting, and trained an Ultralytics YOLO model using PyTorch.
- Evaluated predictions on unseen video frames and generated detection outputs for real-time surveillance scenarios.
- Built a reproducible Python workflow using OpenCV, NumPy, TorchVision, and Ultralytics for dataset preparation and model evaluation.

Visualization and Data Mining Project | Python, Pandas, NumPy, Matplotlib, Seaborn

- Developed a data visualization and analysis workflow to extract insights from large datasets through systematic data exploration.
- Performed data cleaning, feature selection, exploratory analysis, and visualization to identify patterns and correlations.
- Used Python, Pandas, NumPy, Matplotlib, and Seaborn to communicate analytical findings and support data-driven decision-making.

RAG Chatbot for Immigration Services | Python, LangChain, Databricks, MLflow, Meta Llama 3.3 70B

- Built a production-oriented Retrieval-Augmented Generation chatbot using a document-based knowledge base to answer study permit and immigration questions.
- Implemented Databricks Vector Search and LangChain for semantic retrieval and context-aware response generation.
- Integrated Meta Llama 3.3 70B through Databricks Model Serving and developed a Next.js/React frontend deployed on Vercel.
- Used MLflow and Databricks components to support the model development and deployment workflow.

Python Chatbot for Mobile Specifications | Python, Machine Learning, NLP

- Built a Python-based chatbot providing mobile-device specifications through data-driven responses.
- Applied machine learning and natural language processing techniques for query handling.

Learning Management System | Progress 4GL, JavaScript, AJAX

- Created a responsive LMS platform for assignment uploads/downloads, subject assignment, and class-schedule management.

Available for a 12-month January 2027 internship.